EXECUTIVE ANALYTICS CAPSTONE

Customer Churn Analysis

Executive findings, recommended actions, and measurement plan

Sample size	Observed churn rate	Average tenure
15 customers	46.67%	18.8 months

Decision focus

Investigate whether month-to-month customers and customers with repeated support tickets need additional retention support. The dataset is small, so treat these patterns as signals to validate—not proof of cause or a reliable forecast.

1. Executive Summary

The sample contains 15 customers, and 46.67% are marked as churned. Within this small sample, all month-to-month customers and all customers with three or more support tickets are marked as churned. No churn is observed in the one-year or two-year contract groups, or among customers with fewer than three support tickets. These patterns identify useful areas for further review, but the small groups make the rates unstable and prevent broad conclusions.

2. Five Key Insights

Insight	Observed result	Business interpretation
Overall churn	46.67% (7 of 15)	Review preventable departures; validate with a larger sample.
Month-to-month contracts	100% observed churn	Investigate customer reasons and plan fit before offering alternatives.
One-year / two-year contracts	0% observed churn in each group	Potentially useful comparison, but group counts are small.
Support-ticket volume	100% churn with 3+ tickets; 0% with fewer	Audit repeated issues and improve follow-up.
Customer profile	18.8 months average tenure; 84.66 average monthly charges	Descriptive context only; not revenue at risk.

3. Recommended Actions

- **Review month-to-month customers:** ask about reasons for leaving and offer suitable contract options without pressure.
- **Investigate repeated support issues:** review ticket history, unresolved cases, and response/resolution times.
- **Collect exit feedback:** use a short survey to understand whether price, service, or another issue contributed to churn.
- **Monitor the dashboard:** track churn by contract type and support-ticket volume as new data arrives.
- **Run a small pilot:** test targeted follow-up with a defined group before expanding the intervention.

4. Expected Impact and Measurement

The intended impact is better prioritization of retention outreach and fewer preventable customer departures. The current sample does not support a defensible numeric estimate of customers saved or revenue recovered.

How to quantify impact: choose a pilot group, record the number contacted, measure retention after a defined period, and compare results with a suitable comparison group or a clearly documented baseline. Estimate financial impact only when actual revenue/contribution and retention assumptions are available.

Assumption: the observed sample labels and churn definitions are correct. Any future pilot should confirm data quality and use a sufficiently large sample.

5. Limitations and Responsible Interpretation

- Only 15 customers are included; percentages can change sharply when a single customer changes status.
- Group-level rates do not establish that contract type or support tickets cause churn.
- The analysis is descriptive and rule-based; no validated predictive model or controlled experiment was performed.
- Monthly charges are not equivalent to revenue at risk. A credible estimate would require customer-level revenue and explicit retention assumptions.
- Findings should be rechecked on a larger, more representative dataset before major business decisions.

6. Conclusion

Start by investigating month-to-month customers and customers with repeated support tickets. Use the dashboard to monitor trends, validate the patterns on a larger dataset, and measure the impact of any retention action before scaling it.

Data source: supplied customer churn sample CSV. Values are descriptive statistics from this 15-customer sample.